Notes: Kuchler & Zafar (JF, 2019)

Personal Experiences and Expectations about Aggregate Outcomes

Contents

1 Big picture

- **Question.** Do local or personal experiences affect expectations about aggregate outcomes?
- **Main result.** Recent local house price changes affect expected U.S. house price growth, while own unemployment affects expected national unemployment.
- **Mechanism.** The effect is not tied to objective informativeness and is stronger among less sophisticated respondents, pointing toward naive extrapolation rather than optimal limited-information updating.
- **Domain specificity.** Housing experiences mostly affect housing expectations; labor experiences mostly affect labor-market expectations.

2 Incremental contribution of the paper

- It moves the experience-effects literature from lifetime aggregate experiences to recent local and truly personal experiences.
- It combines geographic variation in ZIP/MSA/state house price histories with within-person variation in employment transitions.
- It documents both first-moment extrapolation and second-moment extrapolation from experienced volatility.
- It directly tests whether local experience use is rational by comparing extrapolation to objective informativeness.
- It shows that sophistication moderates extrapolation and that experience effects are domain-specific.

3 Data and measurement

3.1 Survey of Consumer Expectations

The paper uses the Federal Reserve Bank of New York Survey of Consumer Expectations from December 2012 to April 2017. The survey is a monthly rotating panel of roughly 1,200 household heads. Respondents participate for up to 12 months. The final sample has 8,104 respondents age 25 or older with required expectations and demographics.

Interpretation. The panel and ZIP-code information make it possible to connect individual beliefs to both local housing histories and within-person labor-market shocks.

3.2 House price expectations and experiences

Respondents report expected U.S. house price changes at one- and two-year horizons and a subjective distribution over future house price growth. Local experiences are measured using ZIP-code, MSA, and state house price changes going back to 1976. Year-over-year monthly changes remove seasonal effects.

Interpretation. The data allow the paper to study both expected levels and perceived uncertainty.

3.3 Employment expectations and own labor shocks

Respondents report current employment status and their perceived probability that U.S. unemployment will rise over the next year. The data contain 271 job-loss transitions and 323 job-finding transitions.

Interpretation. These transitions allow fixed-effect comparisons of the same respondent before and after a labor-market shock.

4 Models and empirical specifications

4.1 Baseline experience-expectation regression

$$Expectation_{it}^{d} = \alpha + \beta Experience_{it}^{d} + \delta' X_{it} + \gamma_t + \epsilon_{it}.$$

Interpretation. The coefficient β measures how personal or local experiences enter aggregate beliefs after common monthly information is absorbed by time fixed effects.

4.2 House price specification

$$E_{it}(\Delta HP^{US}) = \alpha + \beta \Delta HP_{g(i),t-1} + \delta' X_i + \gamma_t + \epsilon_i,$$

where $g(i)$ is ZIP code, MSA, or state.

Interpretation. A nonzero coefficient rejects the strongest full-information benchmark because all respondents are forecasting the same national outcome.

4.3 Weighted local experience

$$Experience_i(\lambda, H) = \sum_{s=1}^{H} w_s(\lambda, H) \Delta HP_{g(i),t-s}.$$

Interpretation. The best-fitting histories place high weight on recent local house price movements.

4.4 Employment fixed-effect specification

$$Pr_i(U.S.unemploymenthigher) = \alpha_i + \gamma_t + \beta EmploymentStatus_{it} + \epsilon_{it}.$$

Interpretation. Individual fixed effects hold constant stable respondent pessimism and background traits.

5 Identification strategy

- **Local housing design.** Respondents face different local histories but the same national aggregate, with month fixed effects absorbing common news.
- **Informativeness test.** If local experience were used optimally, it should matter more where it historically predicts the national aggregate better. The paper does not find this.
- **Employment design.** Within-person labor transitions identify how the same person's aggregate unemployment beliefs change after job loss or job finding.
- **Heterogeneity.** Stronger effects among less numerate and non-college respondents support a behavioral interpretation.
- **Domain specificity.** Experiences do not broadly spill over to unrelated macro expectations.

6 Tables and figures

6.1 Table I: Summary Statistics

Read: The sample contains 8,104 respondents; average expected house price growth is 5.46%; expected house price uncertainty averages 2.75%; ZIP-level price data are available for 6,032 respondents.

Interpretation. The data provide the cross-sectional and panel variation needed for the paper.

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Table I
Summary Statistics

The table reports summary statistics of the characteristics, house price expectations, and past house price experience of respondents to the Survey of Consumer Expectations (SCE) used throughout the paper.

	N	Mean	*SD*	25th Percentile	50th Percentile	75th Percentile
Respondent Characteristics						
Age	8,104	51.00	14.38	39	52	62
White	8,104	0.85	0.35	1	1	1
Black	8,104	0.09	0.28	0	0	0
Male	8,104	0.54	0.50	0	1	1
Married	8,104	0.68	0.47	0	1	1
College (at beginning of sample)	8,104	0.55	0.50	0	1	1
Income	8,104	80,694	52,784	45,000	67,500	125,000
Numeracy score (% correctly answered)	7,695	0.80	0.22	0.60	0.80	1.00
Homeowner	8,104	0.76	0.43	1	1	1
Years lived in current ZIP	8,099	11.88	11.27	3	8	17
Years lived in current state	8,100	34.69	19.98	18	34	50
Expected House Price Changes						
Expected house price change (point estimate)	8,104	5.46	8.65	2.00	5.00	8.00
Expected std of house price change	7,835	2.75	2.62	1.13	1.78	3.42
Expected change of house price change > 12%	7,866	8.52	20.50	0.00	0.00	5.00
Expected house price change in two years	7,907	5.39	7.30	2.00	5.00	8.00
Past House Price Experience—Last year's hp change						
Most recent yearly return in ZIP code	6,032	5.95	5.44	2.50	5.57	8.89
Most recent yearly return in MSA	6,925	5.56	4.00	2.96	5.03	7.44
Most recent yearly return in State	8,104	5.77	3.40	3.46	5.44	7.14
Unemployment Experience						
Transitions from employment to unemployment	8,104	0.03	0.20	0	0	0
Transitions from unemployment to employment	8,104	0.04	0.22	0	0	0
Unemployment Expectations						
Expected likelihood of higher unemployment						
All	8,104	36.54	23.05	20.00	35.00	50.00

(*Continued*)

(a) Part 1

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Table I—*Continued*

	N	Mean	*SD*	25th Percentile	50th Percentile	75th Percentile
Only employed	5,686	36.94	22.79	20.00	36.00	50.00
Only unemployed	329	43.50	25.39	20.00	47.00	60.00
Local Unemployment rate	8,104	5.59	1.97	4.20	5.20	6.60
Own employment expectations of employed						
Likelihood of job loss if employed	4,896	14.75	20.36	1.00	6.00	20.00
Would find new job within 3 months if lost job	4,973	51.94	32.48	20.00	50.00	80.00
Diff. btw likelihood of	4.896	22.08	27.34	5.00	20.00	40.00

(b) Part 2

6.2 Figure 1: Local House Price Experience and National House Price Expectation; Table II: Previous Year's House Price Change and House Price Expectations

Read: Local house price growth is positively related to expected U.S. house price growth; a 1 percentage point higher local price change raises national expectations by roughly 0.1–0.2 percentage points.

Interpretation. Local experiences influence beliefs about national outcomes even when everyone is forecasting the same U.S. market.

Panel A. ZIP-Level House Price Change

Panel B. State-Level House Price Change

Figure 1. Local house price experience and national house price expectation. The figure shows the relationship between local house price changes in the prior year and expected national house price changes in the next year. For each decile of past price changes in the respondent's ZIP code, Panel A shows the average past house price changes and the average expected national house price changes. Panel B shows the equivalent for each state. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 1

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Table II

Previous Year's House Price Change and House Price Expectations

The table shows regression estimates of equation (1). The dependent variable is the expected change in house prices in percentage points as stated by the respondent. Past local house price change is the year-over-year change in the ZIP code (column (1)), MSA (column (2)), or state (column (3)) in which the respondent lives. Standard errors are clustered at the state level. Time fixed effects are included for each survey month. Demographics include indicators for household income categories, respondents' age and age squared, and indicators for employment status and whether respondents own their home, are male, married, went to college, and are white or black. For the "Effect of 1 std when weighted," we use weights based on the American Community Survey (ACS) to make our sample representative of the U.S. population with respect to income, age, education, and region. Significance levels: *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

	(1) ZIP	(2) MSA	(3) State
Panel A: Expected One-Year Change in U.S. House Prices			
Past Local House Price Change	0.095***	0.172***	0.217***
	(0.0181)	(0.0332)	(0.0412)
Time Fixed Effects	Y	Y	Y
Demographics	Y	Y	Y
Effect of 1 std	0.516	0.686	0.738
Effect of 1 std when weighted	0.635	0.838	0.809
Number of observations	6,032	6,925	8,104
R^2	0.0436	0.0388	0.0367

(b) Table II

6.3 Table III and Figure 2: House Price Changes and Expectations by Magnitude and Informativeness

Read: Extrapolation is not systematically stronger where local prices are objectively more informative about national prices.

Interpretation. This weakens the optimal limited-information explanation.

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Table III

House Price Changes and Expectations by Magnitude of Effect in the Data

The table shows regression estimates of equation (1). Standard errors are clustered at the state level. The dependent variable is the expected change in year-ahead house prices in percentage points as stated by the respondent. Past local house price change is the year-over-year change in the ZIP code (column (1)), MSA (column (2)), or state (column (3)) in which the respondent lives. Time fixed effects are included for each survey month. Demographics include indicators for household income categories, respondents' age and age squared, and indicators for employment status and whether respondents own their home, are male, married, went to college, and are white or black. Respondents are grouped into terciles based on the coefficient on prior local house price changes in a regression of national house price changes on prior-year local price changes in the years since availability of our house price data. Significance levels: *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

	Expected One-Year Change in U.S. House Prices		
	(1) ZIP	(2) MSA	(3) State
Local house price change * Low comovement with U.S. house prices	0.0760*	0.182***	0.163**
	(0.0391)	(0.0425)	(0.0626)
Local house price change * Medium comovement with U.S. house prices	0.135***	0.156***	0.195***
	(0.0456)	(0.0386)	(0.0532)
Local house price change * High comovement with U.S. house prices	0.0173	0.108	0.161*
	(0.0310)	(0.0787)	(0.0839)
Time Fixed Effects	Y	Y	Y
Demographics	Y	Y	Y
Low vs. high comovement	−0.0587	−0.0747	−0.00150
	(0.0482)	(0.0887)	(0.0877)
Number of observations	5,163	5,911	6,945
R^2	0.0447	0.0413	0.0374

(a) Table III

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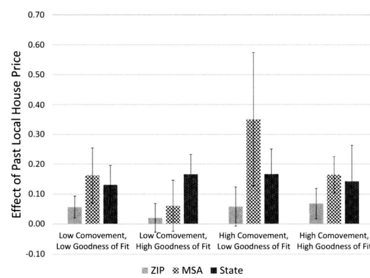

(b) Figure 2

6.4 Figures 3–5 and Table IV: Weighted Experience

Read: Best-fit experience measures place high weight on recent local price changes.

Interpretation. Expectations are consistent with recent-history extrapolation rather than full rational aggregation.

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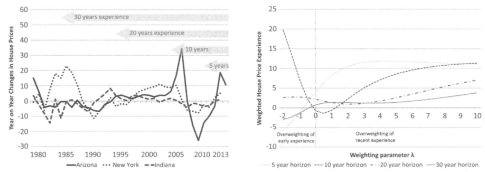

(a) House Price Changes Over Time

(b) Arizona - Weighted Experience

(c) New York - Weighted Experience

(d) Indiana - Weighted Experience

Figure 3. House prices and weighted experience. Panel A shows yearly changes in house prices in Arizona, Indiana, and New York. The remaining panels show how the weighted house price experience of respondents with experience horizons of 5, 10, 20, and 30 years in Arizona (Panel B), New York (Panel C), and Indiana (Panel D) changes as the weighting parameter λ changes. The weighting parameter λ determines the weighting of past changes according to equation (2). (Color figure can be viewed at wileyonlinelibrary.com)

Figure 3 illustrates how geographic variation in house prices translates into the weighted experience variable depending on the weighting parameter λ. Panel A shows yearly changes in house prices in three states with different house price dynamics—Arizona, New York, and Indiana. Arizona experienced high increases in house prices in the early 2000s and a large decline after the onset of the financial crisis in 2008. New York experienced large increases in house prices in the 1980s. Prices also increased in the early 2000s and declined afterwards, though both the increase and subsequent decline of house prices were substantially smaller than in Arizona. House prices in Indiana have been relatively stable over recent decades. As a result, the weighted house price experience in Indiana, reported in Panel D, is very similar for respondents of all experience horizons (irrespective of whether recent or earlier experiences are weighted more). In Arizona and New York, however, weighted experience varies substantially with experience horizon and the weighting parameter λ.

(a) Figure 3

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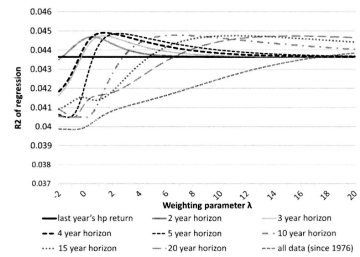
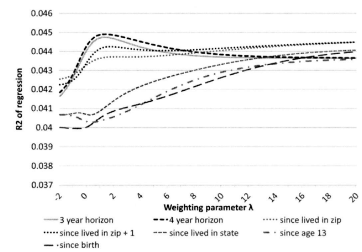


Panel A. Fixed-Year Horizons (All Respondents)



Panel B. Individual-Specific Horizons (Respondents with Years Lived in ZIP Available)

Figure 4. Weighted average of ZIP code house prices and expectation. For each horizon, the figure shows how the R^2 of the regression estimates of equation (1) changes as the weighting

(b) Figure 4

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Table IV
Best-Fit Parameters for Weighted ZIP Code Average as Measure of Experience

For each horizon, the table shows the parameters of the specifications with the highest R^2 in equation (1), where past house price experiences are measured by a weighted average of past house price changes according to equation (2). The estimated parameters shown are the R^2 of the regression, the corresponding estimate of λ, the coefficient on the experience variable, its standard error, and the effect of one standard deviation of the experience variable on the expected house price change.

	Best Fit Parameters for Weighted Past Experiences				
Horizon	R^2 (1)	λ (2)	Coefficient (3)	Standard Error of Coefficient (4)	Effect of 1 Standard Deviation (5)
2 years	4.470%	0.5	0.136	0.018	0.626
3 years	4.475%	1.3	0.149	0.022	0.638
4 years	4.490%	1.4	0.165	0.022	0.668
5 years	4.487%	2.5	0.160	0.025	0.654
10 years	4.478%	6.8	0.158	0.023	0.641
15 years	4.475%	11.1	0.157	0.023	0.636
20 years	4.474%	15.3	0.156	0.023	0.634
All data	4.385%	20.0	0.178	0.028	0.642
Number of Individuals	6,032				

(a) Table IV

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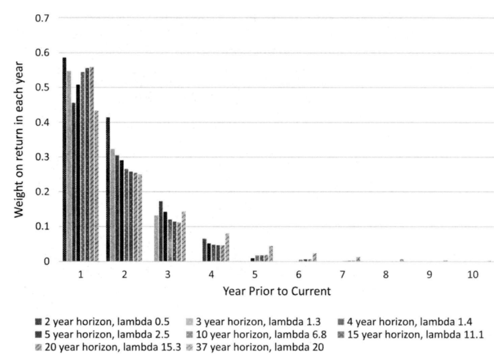

Figure 5. Weights implied by optimal weighting parameter—ZIP code house prices. The

(b) Figure 5

6.5 Table V: Weighted Experience Heterogeneity; Table VI: Past Variation and Expected Variation

Read: Weighted experience effects survive heterogeneity splits, and experienced local volatility increases perceived national house price uncertainty.

Interpretation. The results extend experience effects to both means and subjective uncertainty.

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Table V

House Price Experiences and Expectations by Magnitude of Effect in the Data

The table shows regression estimates of equation (1). Standard errors are clustered at the state level. The dependent variable is the expected change in one-year-ahead house prices in percentage points as stated by the respondent. Past house price experiences are measured by a weighted average of past house price changes according to equation (2). Time fixed effects are included for each survey month. Demographics include indicators for household income categories, respondents' age and age squared, and indicators for employment status and whether respondents own their home, are male, married, went to college, and are white or black. Respondents are grouped into terciles based on the coefficient on prior local house price changes in a regression of national house price changes on prior-year local price changes in the years since availability of our house price data. For each horizon, past house prices are weighted using the λ corresponding to the specifications with the highest R^2 as shown in Table IV. Significance levels: *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

	Expected One-Year Change in U.S. House Prices: Weighted Past House Price Changes Over				
	2 Years (1)	3 Years (2)	5 Years (3)	10 Years (4)	15 Years (5)
Past house price changes * Low comovement with U.S. house prices	0.157***	0.147***	0.171***	0.170***	0.173***
	(0.0325)	(0.0349)	(0.0352)	(0.0331)	(0.0322)
Past house price changes * Medium comovement with U.S. house prices	0.114***	0.140***	0.130***	0.136***	0.132***
	(0.0349)	(0.0408)	(0.0459)	(0.0471)	(0.0427)
Past house price changes * High comovement with U.S.	0.137***	0.159***	0.158***	0.146***	0.158***
	(0.0376)	(0.0379)	(0.0545)	(0.0416)	(0.0500)

(a) Table V

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Table VI

Past Variation in House Price and Expected Variation

The table shows regression estimates of equation (1). The dependent variable is the standard deviation of expected change in one-year-ahead house prices in percentage points as stated by the respondent. For each horizon, the table shows the estimated coefficient on the standard deviation of experienced changes. The standard deviation of past house price changes is based on house prices in the ZIP code (column (1)), MSA (column (2)), and state (column (3)) in which the respondent lives. Standard errors are clustered at the state level. Time fixed effects are included for each survey month. Demographics include indicators for household income categories, respondents' age and age squared, and indicators for employment status and whether respondents own their home, are male, married, went to college, and are white or black. Significance levels: *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

	Std of Expected House Price Change		
	ZIP (1)	MSA (2)	State (3)
Std of house price changes since			
5 years ago	0.0489***	0.0125	0.0268
	(0.0159)	(0.0133)	(0.0218)
10 years ago	0.0307***	0.0236**	0.0143
	(0.0103)	(0.0109)	(0.0104)
20 years ago	0.0370***	0.0250**	0.0179
	(0.00847)	(0.0101)	(0.0109)
1976 (all available data)	0.0391***	0.0168	0.0157

(b) Table VI

6.6 Table VII: Effect of Employment Status on Unemployment Expectations

Read: Own unemployment increases pessimism about national unemployment in within-person specifications.

Interpretation. People treat personal labor-market shocks as information about aggregate labor-market risk.

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Table VII

Effect of Employment Status on Unemployment Expectations

The table shows regression estimates of equation (1). Standard errors are clustered at the respondent level. Employment status is each respondent's self-reported current employment status. In columns (5) and (6), employed respondents who were not previously unemployed are classified as *Employed*. Respondents who are looking for work and were not previously employed are classified as *Unemployed*. Respondents who are currently employed but were unemployed in any previous survey module are classified as *Become Employed*. Respondents who are currently looking for work but were employed in any previous survey module are classified as *Become Unemployed*. Local unemployment is the unemployment rate in the ZIP code in which the respondent lives. Time fixed effects are included for each survey month. Demographics include indicators for each of the 11 possible categories of household income. When no individual fixed effects are included, demographics also include respondents' age, age squared, and indicators for whether respondents are male, married, went to college, and are white or black. Significance levels: *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

	Percent Chance U.S. Unemployment Higher in a Year					
	(1)	(2)	(3)	(4)	(5)	(6)
Employment status						
Employed			Omitted			
Unemployed	6.734***	5.514***	1.442**	1.447**		
	(0.942)	(0.948)	(0.655)	(0.655)		
Become employed					1.122	−2.153*
					(1.115)	(1.116)
Become unemployed					5.217***	−0.172
					(1.430)	(0.986)
Always unemployed					5.724***	0.897
					(1.205)	(1.102)
Retired	−3.144***	−2.901***	−0.539	−0.528	−2.892***	−0.876
	(0.495)	(0.641)	(0.750)	(0.749)	(0.642)	(0.769)
Student	2.714	1.782	0.123	0.117	1.822	−0.439
	(2.030)	(2.069)	(1.685)	(1.686)	(2.072)	(1.731)
Out of the labor force	1.953**	0.690	−0.663	−0.663	0.720	−1.082
	(0.889)	(0.914)	(0.990)	(0.988)	(0.915)	(1.003)

6.7 Tables VIII–X: Heterogeneity by Sophistication, Age, Homeownership, and Labor-Market Characteristics

Read: Less numerate and non-college respondents extrapolate more from both housing and unemployment experiences; age and homeownership do not strongly explain the house-price effect.

Interpretation. Sophistication matters more than risk exposure.

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Table VIII

House Price Change and Expectations by Numeracy and College

The table shows estimates of equation (1). Standard errors are clustered at the state level. Past local house price change is the year-over-year change in the ZIP code (column (1)), MSA (column (2)), or state (column (3)) in which the respondent lives. Time fixed effects are included for each survey month. Demographics include indicators for household income categories, respondents' age and age squared, and indicators for employment status and whether respondents own their home, are male, married, went to college, and are white or black. Significance levels: *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

	Expected One-Year Change in U.S. House Prices		
	ZIP (1)	MSA (2)	State (3)
Panel A: By Numeracy			
Past Local House Price Change * Low Numeracy	0.141**	0.273***	0.317***
	(0.0652)	(0.0726)	(0.0777)
Past local house price change * Medium numeracy	0.0951***	0.163***	0.210***
	(0.0276)	(0.0523)	(0.0568)
Past local house price change * High numeracy	0.0563**	0.0990***	0.157***
	(0.0217)	(0.0297)	(0.0417)
Time fixed effects	Y	Y	Y
Demographics	Y	Y	Y
Low vs. high numeracy	−0.0844	−0.174***	−0.160**
	(0.0622)	(0.0638)	(0.0652)
Number of observations	5,752	6,593	7,695
R^2	0.0479	0.0399	0.0377
Panel B: By College			
Past Local House Price Change * College	0.0784***	0.144***	0.181***
	(0.0207)	(0.0305)	(0.0495)
Past local house price change * No college	0.115***	0.202***	0.261***
	(0.0334)	(0.0560)	(0.0465)
Time fixed effects	Y	Y	Y

(a) Table VIII

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Table IX

House Price Change and Expectations by Age and Homeownership

The table shows regression estimates of equation (1). Standard errors are clustered at the state level. The dependent variable is the expected change in one-year-ahead house prices in percentage points as stated by the respondent. Past local house price change is the year-over-year change in the ZIP code (column (1)), MSA (column (2)), or state (column (3)) in which the respondent lives. Time fixed effects are included for each survey month. Demographics include indicators for household income categories, respondents' age and age squared, and indicators for employment status and whether respondents own their home, are male, married, went to college, and are white or black. We also include indicators for each decile of years lived in the current ZIP code. Significance levels: *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

	Expected One-Year Change in U.S. House Prices		
	ZIP (1)	MSA (2)	State (3)
Panel A: By Age			
Past local house price change * Age 25 to 39	0.0497	0.175***	0.157***
	(0.0507)	(0.0534)	(0.0535)
Past local house price change * Age 40 to 49	0.155***	0.207***	0.294***
	(0.0498)	(0.0693)	(0.0844)
Past local house price change * Age 50 to 59	0.0632*	0.183***	0.215***
	(0.0322)	(0.0505)	(0.0683)
Past local house price change * Age 60 plus	0.115***	0.145***	0.218***
	(0.0270)	(0.0387)	(0.0428)
Time fixed effects	Y	Y	Y
Demographics	Y	Y	Y
Difference age 60 plus vs. 25 to 39	0.0653	−0.0300	0.0614
	(0.0540)	(0.0413)	(0.0595)
Number of observations	6,028	6,921	8,099
R^2	0.0449	0.0392	0.0377
Panel B: By Homeownership			
Past local house price change *	0.0842***	0.161***	0.198***

(b) Table IX

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Table X

Effect of Unemployment on Expectations by Respondent Characteristics

The table shows estimates of equation (1) with *Looking for work* interacted with numeracy, college, and age. Standard errors are clustered at the respondent level. Significance levels: *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

	Percent Chance U.S. Unemployment Higher in a Year		
	(1)	(2)	(3)
Employed		(omitted)	
Looking for work * Low numeracy	5.092**		
	(2.368)		
Looking for work * Medium numeracy	0.222		
	(1.118)		
Looking for work * High numeracy	−0.494		
	(1.217)		
Looking for work * No college		2.412**	
		(1.151)	
Looking for work * College		0.400	
		(0.971)	
Looking for work * Age under 35			1.626
			(1.820)
Looking for work * Age 35 to 50			2.106
			(1.339)
Looking for work * Age 50 to 65			0.900
			(1.060)
Retired	0.772	1.542	1.315
	(2.177)	(2.097)	(2.135)
Student	−5.367	−4.891	−4.980
	(3.727)	(3.897)	(3.858)
Out of the labor force	1.170	3.434	3.188
	(2.584)	(3.146)	(3.141)
Time fixed effects	Y	Y	Y
Demographics	Y	Y	Y
Individual fixed effects	Y	Y	Y

Table X

6.8 Table XI: Own Experiences and Expectations about Other Economic Outcomes

Read: Experiences mostly affect expectations in the same domain and do not broadly spill over to interest rates, stocks, inflation, or government debt.

Interpretation. This supports domain-specific extrapolation rather than general pessimism.

Table XI

Own Experiences and Expectations about Other Economic Outcomes

The table shows estimates of equation (1) with the following dependent variables: whether the respondent believes that they will be better off a year later (column (1)) and are better off than a year ago (column (2)), both on a five-point scale; the percentage chance that interest rates on saving accounts (column (3)) and U.S. stock prices (column (4)) will be higher a year later; expected inflation in the next year (column (5)) and between two and three years from now (column (6)); the expected change in government debt (column (7)) and U.S. home prices (column (8)); and likelihood of higher unemployment (column (9)). Fixed effects are included for each month. Demographics include indicators for household income categories, respondents' age and age squared, and indicators for employment status and whether respondents own their home, are male, married, went to college, and are white or black. Controls also include past house price changes in Panel A, column (8). Standard errors are clustered at the state level. Significance levels: *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

			Percentage Chance That the Following Will Be Higher in a Year						
	Will Be Better Off in a Year	*Are* Better Off than Year Ago	Interest Rates on Savings	U.S. stock Prices	Inflation 1 Year	Inflation 3 Years	Government Debt	Home Prices	Unemploy-ment
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Effect of Own Employment Status									
Employment status									
Employed	(omitted)					(omitted)			
Looking for work	−0.0573**	−0.476***	−0.253	−0.876	0.113	0.0802	45.60	−1.107**	
	(0.0265)	(0.0380)	(0.677)	(0.640)	(0.111)	(0.115)	(48.74)	(0.514)	
Retired	−0.0618**	−0.222***	0.332	−0.627	−0.000399	−0.175	−236.4	−0.589	
	(0.0273)	(0.0307)	(0.833)	(0.751)	(0.111)	(0.124)	(236.2)	(0.381)	
Student	−0.0800	−0.362***	−3.410*	−3.353**	0.0316	−0.112	64.44	1.019	
	(0.0719)	(0.0760)	(1.909)	(1.455)	(0.263)	(0.274)	(68.24)	(0.911)	
Out of labor force	−0.126***	−0.277***	−1.166	−1.895**	−0.0814	−0.143	537.4	−0.0235	
	(0.0329)	(0.0397)	(1.057)	(0.877)	(0.162)	(0.191)	(533.8)	(0.626)	
Local unemployment indicators	Y	Y	Y	Y	Y	Y	Y	Y	
Demographics	Y	Y	Y	Y	Y	Y	Y	Y	
Time fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	
Individual fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	
Number of observations	60,672	60,663	60,668	60,147	52,888	52,881	52,744	45,281	
Number of individuals	8,104	8,103	8,104	8,104	7,770	7,752	7,921	6,038	

(Continued)

Table XI—*Continued*

Panel B: Effect of Local House Price Change								
ıl house price change	0.00159	0.000350	−0.00896	0.00205	0.00828	−0.00296	−0.0523	−0.062
de level)	(0.00194)	(0.00224)	(0.0679)	(0.0744)	(0.0337)	(0.0379)	(0.0385)	(0.0610
ːmployment indicators	Y	Y	Y	Y	Y	Y	Y	Y
phics	Y	Y	Y	Y	Y	Y	Y	Y
d effects	Y	Y	Y	Y	Y	Y	Y	Y
ɔf observations/ individuals	6,030	6,025	6,029	5,985	6,003	6,003	5,881	6,032

6.9 Tables XII–XIII: Expectations and Outcomes

Read: Job-loss expectations predict realized job loss, and housing expectations predict whether respondents view local home investment positively.

Interpretation. Survey expectations are economically meaningful because they relate to realized outcomes and attitudes.

Table XII

Predictiveness of Own Employment Prospects

The table shows regression estimates for whether respondents' self-reported probability of losing their job is indicative of future job loss. The dependent variable is whether respondents report having lost their job within the next one, three, six, or nine months of the survey module. Pr(*job loss within 12 months)* is the percentage chance that the respondent will lose her job within the next 12 months as stated by the respondent (on a zero to one scale). Panel A includes only the first survey module for each respondent. Panel B includes all survey modules and respondent fixed effects. Local unemployment is the unemployment rate in the ZIP code the respondent lives in. Time fixed effects are included for each survey month. In all specifications, demographics include indicators for each of the 11 possible categories eliciting household income in the survey. When no individual fixed effects are included, demographics also include respondents' age and age squared, and indicators for whether respondents are male, married, went to college, and are white or black. Standard errors are clustered at the respondent level when applicable. Significance levels: *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

Panel A: Employment Prospects When Entering Sample				
	Lose Job Within			
	One Month (1)	Three Months (2)	Six Months (3)	Nine Months (4)
Pr(job loss within 12 months)	0.0359***	0.0893***	0.140***	0.163***
	(0.0114)	(0.0174)	(0.0219)	(0.0239)
Local unemployment (Indicators for decile)	Y	Y	Y	Y
Demographics	Y	Y	Y	Y
Time fixed effects	Y	Y	Y	Y
Number of observations	4,865	4,696	4,463	4,225

Panel B: Within Individual Changes in Employment Prospects				
	Lose Job Within			
	One Month (1)	Three Months (2)	Six Months (3)	Nine Months (4)
Pr(job loss within 12 months)	0.0193***	0.0178*	0.0158	0.0134
	(0.00563)	(0.00922)	(0.0100)	(0.00963)

(a) Table XII

Table XIII

House Price Expectations and Housing Investment

The table shows ordered logit regression estimates of the effect of expected house price changes on how attractive respondents consider investing in a home in their current ZIP code. Respondents can choose whether they consider such an investment to be a bad or very bad investment, neither a bad nor good investment, a good investment, or a very good investment. Demographics include indicators for household income categories, respondents' age and age squared, and indicators for employment status and whether respondents own their home, are male, married, went to college, and are white or black. Significance levels: *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

	Housing in ZIP is a Good Investment		
	One-Year National Expectations (1)	One-Year ZIP Expectations (2)	Five-Year ZIP Expectations (3)
Expectations—first tercile		(omitted)	
Expectations—second tercile	0.290***	0.625***	0.400***
	(0.0709)	(0.0728)	(0.0752)
Expectations—third tercile	0.230***	0.754***	0.593***
	(0.0773)	(0.0804)	(0.0767)

(b) Table XIII

7 Interpretive synthesis

Experience-based expectations are local, domain-specific, and stronger among less sophisticated respondents. These facts suggest that belief heterogeneity in macro and housing markets can arise from personal histories rather than only from public information or preferences.

8 Short summary

- Local house price changes affect expectations about national house prices.
- Local house price volatility affects perceived uncertainty about national house prices.
- Personal unemployment affects expectations about national unemployment.
- Experience effects are not proportional to objective informativeness.
- Less sophisticated respondents extrapolate more.
- Experiences mostly affect expectations in the same domain.
- Expectations predict future job loss and housing investment attitudes.

9 Conclusion

9.1 Core conclusion

Recent personal experiences affect expectations about aggregate outcomes.

9.2 Incremental contribution

The paper provides evidence on local and personal experience, first and second moments, heterogeneity by sophistication, domain specificity, and links from expectations to behavior.

9.3 Takeaway

Aggregate belief formation is heterogeneous because individuals carry different local and personal histories into their forecasts.